Teaching a Small Model to Deliberate

Verifier-Anchored Reinforcement Learning over a Frozen LLM

2026

Motivation

The pain: cheap models, expensive reasoning

- Small 1–7B models are fast and deployable, but lag on GSM8K and MATH
- The gap is widest exactly where answers are checkable by a non-LLM verifier
- Test-time search (best-of-N, self-consistency, MCTS) helps, but is amnesiac: every query pays full cost again, nothing is distilled back
- The question: can we close the gap *without growing the model*?

And learned rewards get gamed

- Dense *learned* reward models are seductive but open a Goodhart attack surface
- Under enough optimization pressure, the policy converges on the proxy's defects
- Our anchor: a deterministic non-LLM verifier (SymPy equality, unit tests, Lean type-check)
- A signal enters the reward channel only if it is an objective proposition a program can check

Thesis

1. Freeze the base LLM; learn small heads π_θ, Q_ψ over a structured state
2. Cast LLM deliberation on verifiable tasks as approximate Q-learning over an induced context MDP
3. Reward is verifier-anchored, not a learned critic
4. MCTS is a sample-efficient *policy-improvement operator* that supplies stronger training targets, not the central algorithm

The setup: an induced context MDP

The induced context MDP

- State s: a structured reasoning context (not raw token soup)
- Actions A: seven coarse cognitive verbs, $|A| = 7$
- Transition P: frozen LLM + prompt template samples the next state; non-enumerable
- Reward R: ground-truth terminal $+1/-1$ from the verifier; small capped intermediate shaping
- Controller observes only *sampled* transitions: model-free RL with the LLM as the environment

Structured state: a shell the LLM fills

- Slots: problem P, Facts, open/closed Subgoals, Hypotheses, verifier log, committed answer, depth
- Load-bearing split: **Fact** ("I have proven this") vs **Hypothesis** ("I think this is true")
- Avoids two failure modes: raw tokens (no inductive bias) and one opaque embedding (no action masking)
- Serialized with structural markers: `[FACT]`, `[HYPOTHESIS]`, `[VERIFIER_CALL]`, `[STATE_END]`

Actions: seven cognitive verbs

$A = \{$ `DECOMPOSE`, `COMPUTE_SUBEXPR`, `FORM_HYPOTHESIS`, `CHECK_CASE`, `INVOKE_VERIFIER`, `BACKTRACK`, `COMMIT_ANSWER` $\}$

- Only the *verb* is categorical; the LLM verb-instantiator generates the target id and content
- Action space stays cardinality-7 regardless of tree depth; a state-conditioned mask gates illegal verbs
- `INVOKE_VERIFIER` is a first-class budgeted move, not a post-hoc parser pass

A concrete trace (GSM8K)

P: "A store sold 7 boxes with 6 pencils each. How many pencils?"

1. `DECOMPOSE` `sg_root`, then `FORM_HYPOTHESIS`: `hyp_1` = "the total is 42" (tentative)
2. `COMPUTE_SUBEXPR` "7×6" $\rightarrow$ 42, added as a derived Fact
3. `INVOKE_VERIFIER` on `hyp_1` (SymPy) $\rightarrow$ confirmed; `hyp_1` promoted to Fact ($+0.05$, capped)
4. `COMMIT_ANSWER` "42" $\rightarrow$ terminal verifier check $\rightarrow$ reward $+1$

The learner

Architecture: frozen LLM, tiny trainable heads

- One frozen base LLM is *both* encoder and stochastic transition/generator (one forward, not two)
- State embedding = hidden state at `[STATE_END]`; a small adapter MLP
- Multi-head $Q_\psi(s, a)$ over the 7 verbs, plus a policy-prior head π_θ sharing the adapter
- Total trainable params under 5M: 2–3 orders of magnitude smaller than the frozen LLM
- Approximate Q-learning: replay buffer, target net synced every N_{sync}, prioritized by TD error

Offline by design: learn from any trajectory

- Q-learning is off-policy: the max bootstrap refers to the greedy continuation, not the behavior policy
- So the learner can train from *any* trajectories covering the state-action space
- Sources: its own runs, other models' runs, logged production data, human demonstrations
- Decouples data collection from learning: a genuine strength, not a limitation
- Q_{MCTS} targets are stored *with* each replay sample, so stale entries do not break correctness

MCTS as a policy-improvement operator (not the star)

- The heads encode *average* behavior; MCTS spends compute *locally* on this state
- PUCT, with the learned policy as prior and Q_ψ as leaf value:

$$a^\star = \arg\max_a \; \widehat{Q}(s,a) \; + \; c_{\text{puct}}\, \pi_\theta(a \mid s)\, \frac{\sqrt{N(s)}}{1+N(s,a)}$$

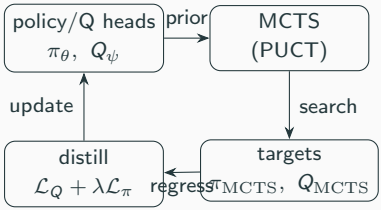
- Decision/chance-node (afterstate) split; the chance-node mean is an honest estimator of the Bellman inner expectation:

$$\frac{W(s,a)}{N(s,a)} \;\approx\; \sum_{s'} P(s' \mid s,a)\,\big[R(s,a,s') + \gamma V(s')\big]$$

Output: stronger targets π_{MCTS} (visit counts) and Q_{MCTS} (chance-node Q).

The self-reinforcing expert-iteration loop

- Distill π_{MCTS} into the policy head ($\mathcal{L}_\pi$, cross-entropy)
- Regress Q_ψ onto Q_{MCTS}, used *in place of* single-sample Bellman targets
- $\mathcal{L}_{\mathrm{total}} = \underbrace{(Q_\psi - Q_{\mathrm{MCTS}})^2}_{\mathcal{L}_Q} + \lambda\, \mathcal{L}_\pi$
- Fast vs slow: the heads are System 1, MCTS is System 2; training shifts competence into System 1



Reward hacking: a verifier is necessary, not sufficient

- Goodhart is the classic RL failure: optimize a mis-specified reward, get gamed behavior that scores well and reasons badly
- A non-LLM verifier removes the *learned*-reward attack surface, but a poorly specified verifier reward is still gameable
- We red-teamed our own intermediate shaping reward and found a farm: verify content-free tautologies or self-authored claims for free reward

Fix by construction, not by patch

Tie the rewarded check to the target claim; reward only the *first* success per target; hard-cap cumulative intermediate reward at $\frac{1}{5}$ of the terminal magnitude $\Rightarrow$ the ground-truth terminal verifier always dominates, so farming is *bounded*, not merely discouraged.

Inference and deployment

The compute knob B: fast by default, think harder on demand

$B = 0$ (fast)

masked arg max over Q_ψ; one LLM forward per decision; constant latency

$B > 0$ (search)

run PUCT MCTS for B simulations; sample the visit-count policy

- Same trained Q_ψ and π_θ in both modes; one dial traces a compute-vs-accuracy frontier, no retraining
- Fast mode is competitive precisely because the heads were distilled from MCTS targets

Online continuous learning in deployment

- Because it is off-policy and feedback-anchored, learning need not stop at deploy
- Any verifier-style signal keeps it improving, including user thumbs up/down
- The signal must stay a *structural check*, never a learned classifier in the Bellman target
- Adapter-tier separation: per-user adapters absorb idiosyncratic feedback; the global adapter accepts only changes that survive aggregation

A model that gets better the longer it is used.

Positioning and evaluation

Positioning and defensible novelty

- **Search-and-learning in games** (AlphaZero, expert iteration): we invert it, agent and Q_ψ central, search optional
- **Verifier-anchored reasoning** (PRMs, AlphaCode, AlphaProof, DeepSeek-R1): the verifier-vs-aesthetic asymmetry is load-bearing; `INVOKE_VERIFIER` is a callable verb
- **VerifierQ**: verifier-as-*action* not verifier-as-Q; online distilled targets not fixed-dataset fitting; categorical multi-head Q
- **rStar-Math**: same family, but ours is a structured-state seven-verb MDP
- **vs STaR / ReST-EM / Math-Shepherd**: we distill per-step Q_{MCTS} as a genuine Bellman target, not trajectory-level or a process-reward model

How we will evaluate (the plan, not results)

- Benchmarks GSM8K and MATH; a baseline grid fixed in advance: base-greedy, best-of-N, self-consistency, STaR/ReST-EM, AlphaZero-for-reasoning, PRM-guided
- Pre-registered falsifiable hypotheses and the headline frontier figure committed *before* any training
- ≥ 3 training seeds, ≥ 8 inference replicates; report mean $\pm$ std on every point
- Holm–Bonferroni across the FLOP $\times$ benchmark grid; explicit threat model for compute leaks

Honest frontier accounting

x: total inference FLOPs (charge *every* LLM forward inside the B simulations, not just the root); y: accuracy. FLOP-matched curves. Pilot found generation dominates prefill $\approx$ 79:1 per decision.

Vision

A small, fast model that learns to deliberate over structured states,
anchored by verifiable (non-learned) rewards.

Offline today, online-continual in deployment,
with a single test-time compute knob.

MCTS earns its keep only as the operator that manufactures stronger targets.
Next: run the pre-registered frontier at scale and report it honestly.